Fair core selections: a sharp boundary between pairs and triples

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Abstract

We compare full leximin and descending-lexicographic-minimum core payoffs in unit-capacity matching markets. For every graph matching game with nonempty core, leximin guarantees half of each player's maximum core payoff and the sharp normalized distance is $\lfloor n/2 \rfloor / (\lfloor n/2 \rfloor + 1)$ for $n \geq 4$. Three-sided assignment games instead permit distance tending to two and no positive dimension-independent individual guarantee, already with weights in $\{0, 1, 2, 3\}$. Complete additive three-sided markets with equal sector sizes retain a one-third guarantee and a sharp distance supremum of $4/3$. The proofs use the entire core and count every player. A priority audit identifies the half-guarantee as a short consequence of existing algorithmic invariants and a classical graph reduction; the sharp distance comparison and sparse triple construction are the principal candidate contributions. External priority and significance are not established.

1 One question, three market classes

The two fairness rules are established objects: leximin raises the lowest payoffs successively, while leximax-min lowers the highest successively. Older egalitarian work and recent combinatorial algorithms motivate their comparison [5, 4]. We ask how much the choice of rule can alter who receives a fixed surplus, and whether leximin protects each player's share of their own best stable outcome.

A unit-capacity matching game has a finite player set N and nonnegative weights on feasible teams. A coalition earns the maximum weight of disjoint teams contained in it. Teams are edges of a graph in the pairwise model, and triples containing one player from each of three sectors in the three-sided model. Let $T > 0$ be the *integral* matching value. The entire core is

$$C = \{p \geq 0 : \ \textstyle\sum_{i \in N} p_i = T, \quad \sum_{i \in e} p_i \geq w_e \text{ for every team } e\}. \tag{1}$$

These inequalities imply all coalition inequalities by summing over an optimal coalition matching. We assume $C \neq \varnothing$. Bipartite graphs ensure this, whereas general graphs and triple arrays need not.

Write $x = \mathrm{Lmin}(C)$ for the lexicographic maximum of ascending sorted payoffs, $z = \mathrm{Lmax}(C)$ for the lexicographic minimum of descending sorted payoffs, and

$$\rho(C) = \frac{\|x - z\|_1}{T}, \qquad H_i(C) = \max_{y \in C} y_i.$$

All players count toward $n = |N|$, including fixed-zero players. These are full selectors, not arbitrary first-stage maximin or minimax points. If $H_i = 0$, any fractional guarantee for that coordinate is vacuous.

The quantity $\rho/2$ has a direct accounting interpretation:

$$\frac{\rho}{2} = \frac{1}{T} \sum_{i:z_i > x_i} (z_i - x_i).$$

It is the minimum fraction of total surplus that must change recipients when moving between the two allocations. Thus the two-sided bound approaches one-half of the surplus, whereas three-way cooperation can require moving almost all of it. This is a worst-case statement, not an empirical claim.

Theorem 1. *For any unit-capacity graph matching game with nonempty core, fix an optimal matching and let $k \geq 1$ count its positive-weight edges. Then*

$$x_i \geq H_i(C)/2 \quad \text{for every } i, \qquad \|x - z\|_1/T \leq k/(k+1).$$

The distance bound holds for any minimax core point z. For $k = 1$ the full selectors coincide. For each $k \geq 2$ a bipartite market attains the bound. Consequently the exact n-player maximum is zero for $n = 2, 3$ and $\lfloor n/2 \rfloor / (\lfloor n/2 \rfloor + 1)$ for $n \geq 4$, both for bipartite markets and for all nonempty-core graph matching games.

Theorem 2. *For all integers $L \geq 1, R \geq 0$, a three-sided assignment game with nonempty entire core and weights in $\{0, 1, 2, 3\}$ has*

$$n = 9(L + R) + 12, \qquad T = 3(L + R + 2),$$

$$\rho(L, R) = \frac{2((R+2)2^L - 1)}{(2^L + 2)(L + R + 2)}. \tag{2}$$

A player has $H_i = 1$ and $x_i = 2/(2^L + 2)$. Hence no positive dimension-independent individual fraction survives, and $R = L^2$ gives $\rho \to 2$, the sharp supremum. The family has equally sized sectors and only $12(L + R) + 9$ positive triples.

Entire-core domain	Uniform x_i/H_i guarantee	Distance supremum
Graph matching, nonempty core	1/2	1
Equal-sized complete additive triples	1/3	4/3
Arbitrary three-sided, nonempty core	none positive	2

The middle row is proved below. It shows why the additive benchmark does not settle the unrestricted question. The graph extension shows that bipartiteness itself is not the transfer mechanism. These statements concern unit capacity and do not assert analogous results for b-matching games.

2 Pairwise transfer and half domination

Summing the inequalities of any optimal positive matching in (1) shows that each matched pair has its designated total s_i and every unmatched player is fixed at zero.

Lemma 1. *For every $y \in C$, $y_i \leq 2x_i$ for every player i. In particular $\|x - y\|_1/T \leq 1$ for every core allocation y.*

Proof. Put $d = y - x$ and suppose $d_a = \delta > 0$. Define global vertex sets

$$A = \{i : d_i \geq \delta\}, \qquad B = \{i : d_i \leq -\delta\}.$$

Matched-pair changes are opposite and unmatched changes are zero, so matching edges biject A and B. At a tight edge uv, feasibility of y gives $d_u + d_v \geq 0$. If $u \in B$ and $v \notin A$, then $d_u + d_v < 0$, a contradiction. Therefore the direction $h = \mathbf{1}_A - \mathbf{1}_B$ preserves the total and harms no tight inequality. Every decreasing coordinate has $x_i \geq \delta$ because $y_i \geq 0$. Finiteness of the slack constraints yields $x + \epsilon h \in C$ for all sufficiently small $\epsilon > 0$.

If $\min_B x > \min_A x$, this step raises the smallest affected payoffs while lowering only strictly larger ones, contradicting full leximin even when smaller fixed payoffs occur elsewhere. Hence

$$\delta \leq \min_B x \leq \min_A x \leq x_a, \qquad y_a = x_a + \delta \leq 2x_a.$$

The other coordinates follow by repeating the argument or by $y_i \leq x_i$. The overlap is at least $T/2$, giving the distance statement. No bipartition was used. □

The fraction is sharp on a single edge. In the bipartite case it also follows from Vazirani's algorithmic invariants; Section 9 makes this priority qualification explicit. The proof above is a direct geometric proof. The extension to nonbipartite nonempty-core games also follows from the classical double-cover reduction, as explained there.

3 The sharp finite bound and its attainment

Proof of Theorem 1: upper bound. On matched pair i, write the change from x to z as $(d_i, -d_i)$, and put

$$q_i = |d_i|, \quad D = \sum_i q_i, \quad M = \max_a z_a > 0.$$

Thus $\|x - z\|_1 = 2D$. Both x payoffs in pair i are at least q_i: the decreasing endpoint by nonnegativity of z, and the increasing endpoint by Lemma 1. The increasing endpoint's z payoff is at least $2q_i$. Therefore

$$s_i \geq 2q_i, \qquad q_i \leq M/2, \qquad D \leq kM/2. \tag{3}$$

These statements also hold when $q_i = 0$.

Some player a satisfies $z_a = M$ and $x_a \geq M$. Otherwise, a sufficiently short move from z toward x lowers every maximum coordinate and leaves every other coordinate below M, contradicting minimax optimality. Since $M > 0$, this player belongs to a positive matched pair p. Its displacement is $q_p = x_a - M$; the other x payoff in that pair is at least q_p. Thus $s_p \geq M + 2q_p$. Summing this strengthened inequality with (3) for the other pairs gives

$$T \geq 2D + M \geq 2D\left(1 + \frac{1}{k}\right).$$

This proves the bound, also when $D = 0$.

If $k = 1$, all other payoffs are fixed at zero. The remaining two sum to T and range over a closed interval of splits. Maximizing their minimum is the same as minimizing their maximum, and uniquely selects the feasible split closest to equality. Hence $x = z$ in this case. □

Attainment and player count. For $k \geq 2$, take a complete $k \times k$ market with identical rows

$$(w_{i1}, \ldots, w_{ik}) = (1, \ldots, 1, 2).$$

Every perfect matching has value $T = k + 1$, and every edge belongs to one. Complementarity consequently makes every edge inequality tight throughout the core. Writing $c = (1, \ldots, 1, 2)$, the entire core is exactly

$$u_i = t \quad (1 \leq i \leq k), \qquad v_j = c_j - t \quad (1 \leq j \leq k), \qquad 0 \leq t \leq 1.$$

Its minimum coordinate is $\min(t, 1 - t)$, uniquely maximized at $t = 1/2$. Its maximum is $2 - t$, uniquely minimized at $t = 1$. Every coordinate changes by $1/2$, so $\|x - z\|_1 = k$ and $\rho = k/(k + 1)$. Adding one isolated player leaves the full selectors otherwise unchanged. Every matching on n players has at most $\lfloor n/2 \rfloor$ edges, which proves the exact n-player formula. □

4 An explicit market

All color indices are modulo three. Introduce global players G_i of color i. At levels $\ell = 0, \ldots, L$ introduce $U_{\ell i}, V_{\ell i}, D_{\ell i}$ for $i = 0, 1, 2$. At even levels their colors are $i, i + 1, i + 2$; at odd levels the colors of U and V are interchanged. Put $s_0 = 2$ and $s_\ell = 1$ for $\ell \geq 1$. Assign the following positive triple values:

Triple	Value
$\{U_{\ell i}, V_{\ell i}, D_{\ell i}\}$	s_ℓ
$\{U_{\ell i}, V_{\ell i}, G_{i+2}\}$	s_ℓ
$\{U_{0i}, G_{i+1}, G_{i+2}\}$	1
$\{U_{\ell i}, V_{\ell-1,j}, V_{\ell-1,k}\}$, $\ell \geq 1$	2
$\{V_{\ell i}, U_{\ell-1,j}, U_{\ell-1,k}\}$, $\ell \geq 1$	3 if $\ell = 1$, else 1

In the last two rows, $\{j, k\} = \{0, 1, 2\} \setminus \{i\}$. Each listed triple has exactly one player of each color.

For every i introduce R copies $(U'_{ri}, V'_{ri}, D'_{ri})$ with the same colors as its top-level triple. Give value 1 to each copy's local and global pair triples $\{U'_{ri}, V'_{ri}, D'_{ri}\}$ and $\{U'_{ri}, V'_{ri}, G_{i+2}\}$, and to its cross-triples

$$\{U'_{ri}, V_{Li}, G_{i+2}\}, \qquad \{V'_{ri}, U_{Li}, G_{i+2}\}.$$

Every unlisted triple has value zero. There are $3(L + R) + 4$ players per sector.

5 Integral value and entire core

The triples using each pair's own D form a disjoint integral matching with weight $6 + 3L + 3R = T$. Consider the allocation assigning 1 to both root U and root V, 0 to every subsequent U, 1 to every subsequent V, the analogous values to copies, and 0 to every dummy. It has total T and covers every weighted triple. Summing its payoffs over any disjoint matching bounds that matching's weight by T. Hence the displayed integral matching is optimal and this allocation belongs to the entire core.

Lemma 2. *Every core allocation, and only a core allocation, has all dummy payoffs zero and the following form:*

$$\begin{aligned} U_{0i} &= 1 + q_{0i}, & V_{0i} &= 1 - q_{0i}, \\ U_{\ell i} &= q_{\ell i}, & V_{\ell i} &= 1 - q_{\ell i} \quad (\ell \geq 1), \\ U'_{ri} &= q_{Li}, & V'_{ri} &= 1 - q_{Li}, \end{aligned}$$

where $0 \leq q_{\ell i} \leq 1$ *and*

$$q_{\ell i} = q_{\ell-1,j} + q_{\ell-1,k}, \qquad q_\ell = (J - I)q_{\ell-1}. \tag{4}$$

Here J *is the* 3×3 *all-ones matrix.*

Proof. Sum the inequalities of the displayed matching. Efficiency and nonnegativity force every unmatched G to zero and every matching triple to equality. The alternative pair triple using G then forces its local D to zero and its pair to total s_ℓ. Root lower-bound triples give $U_{0i} \geq 1$. At level one, the two recurrence inequalities become respectively $q_{1i} \geq q_{0j} + q_{0k}$ and $q_{1i} \leq q_{0j} + q_{0k}$ after substituting the root offsets. At subsequent levels the same substitutions give these opposite inequalities without offsets. Cross-triples give opposite inequalities between a copied U and its original top U, hence equality. Nonnegativity supplies the stated bounds.

Conversely, the parameterization satisfies every displayed triple inequality, has total T, and is nonnegative. All omitted zero-weight triple inequalities are automatic. It therefore satisfies every coalition inequality by summing over a matching. This proves equality with the entire core. □

6 The full selectors and quantitative separation

For minimax, every root U is at least 1. The allocation just used, denoted z, has maximum payoff 1. Any core allocation with maximum at most 1 must have $U_{0i} = 1$ for all i, hence $q_0 = 0$. Lemma 2 then fixes every coordinate. Thus z is the unique minimax allocation and the full descending-lexicographic minimum.

For leximin, all D, G are fixed at zero. Every other coordinate can simultaneously be positive by taking $q_0 = t(1, 1, 1)$ with $0 < t < 2^{-L}$. Consequently the full procedure first fixes exactly the dummy coordinates at zero, then maximizes a common floor η for all remaining coordinates. If such a floor is feasible, level-one U and top-level V imply

$$2\sum_i q_{0i} = \sum_i q_{1i} \geq 3\eta, \qquad 2^L \sum_i q_{0i} = \sum_i q_{Li} \leq 3(1 - \eta).$$

Combining gives $\eta \leq 2/(2^L + 2)$. Equality is attained by

$$t = \frac{1}{2^L + 2}, \qquad q_{\ell i} = 2^\ell t. \tag{5}$$

The least nonroot U is $2t$ and the least V is $1 - 2^L t = 2t$; root and copy payoffs also respect this floor. Equality forces $q_{1i} = \eta$ for all three i. Since $J - I$ is invertible, with eigenvalues $2, -1, -1$, it then forces $q_0 = t(1, 1, 1)$. Thus the optimal positive-floor face is a singleton and (5) gives the full leximin allocation x. No symmetry restriction on the feasible core was imposed.

Every level-ℓ pair changes from z to x by $(2^\ell t, -2^\ell t)$, including the offset root pair. Each copied pair changes by $(2^L t, -2^L t)$. Therefore

$$\|x - z\|_1 = 6t\left(\sum_{\ell=0}^{L} 2^\ell + R2^L\right) = \frac{6((R + 2)2^L - 1)}{2^L + 2}.$$

Dividing by T proves (2). With $R = L^2$, both the amplified payoff fraction and the share of copied pairs tend to one, giving $\rho \to 2$. Conversely, nonnegative vectors of total T always satisfy $\|x - z\|_1 \le 2T$, proving the sharp supremum. For example, $L = 4, R = 16$ gives $\rho = 287/198 > 4/3$. Finally, every top V has core maximum 1, attained at z, but payoff $2/(2^L + 2)$ at x. This proves the individual conclusion separately.

Proposition 1. *Let S_n be the supremum for total player count n, allowing arbitrary sector sizes. For $n \ge 30$, put $k = \lfloor (n-12)/9 \rfloor$. Then*

$$0 \le 2 - S_n \le \frac{2\lceil \log_2 k \rceil + 4}{k+2} + \frac{2}{(k+2)^2} = O(\log n/n).$$

Proof. Choose $L = \lceil \log_2 k \rceil$ and $R = k - L \ge 0$, and add the remaining at most eight players with zero incident values. Matching efficiency forces these players to zero in the entire core, preserving both full selectors. Subtracting (2) from two gives

$$2 - \rho = \frac{2L2^L}{(2^L+2)(k+2)} + \frac{4}{2^L+2} + \frac{2}{(2^L+2)(k+2)}.$$

Use $2^L \ge k$ and bound the first fraction by $2L/(k+2)$. □

7 Additive triples and a separate individual-failure control

Proposition 2. *For a complete three-sided market with m players per sector and additive nonnegative values $w_{ijk} = a_i + b_j + c_k$, full leximin satisfies $x_i \ge H_i/3$ and $\|x - y\|_1/T \le 4/3$ for every core point y. Both constants are sharp across this class.*

Proof. Subtract each sector's minimum from its coefficients. Write the resulting nonnegative offsets as b_{si}, and let S be the sum of the three original minima. Complete-array nonnegativity implies $S \ge 0$. Every perfect matching has the same value, and each triple lies in one, so every triple inequality is tight throughout the core. Subtracting these equalities shows that the entire core is exactly

$$p_{si} = b_{si} + t_s, \qquad t_s \ge 0, \qquad \sum_{s=1}^{3} t_s = S.$$

Here $T = \sum_{s,i} b_{si} + mS$. This additive-core simplex is established structure [10]; its quantitative consequences are immediate. Each sector contains an offset-zero player, so leximin uniquely sets all $t_s = S/3$ (also trivially when $S = 0$). Then $H_{si} = b_{si} + S \le 3x_{si}$ and

$$\|x - y\|_1 = m \sum_s |t_s - S/3| \le 4mS/3 \le 4T/3.$$

An offset-zero player makes the individual fraction sharp when $S > 0$. For $m \ge 2$, the family $w_{ijk} = 1 + \mathbf{1}_{i=1} + \mathbf{1}_{j=1}$ has $T = m + 2$, leximin shifts $(1/3, 1/3, 1/3)$ and minimax shifts $(0, 0, 1)$. Its distance is $4m/[3(m+2)] \to 4/3$. □

Individual loss does not imply large selector disagreement. In the triple recurrence construction, remove copies and root offsets, set every pair total to one, remove root lower-bound triples, and change the level-one reverse weight from three to one. The core is then $U_\ell = (J - I)^\ell U_0$, $V_\ell = 1 - U_\ell$ with all dummy payoffs zero. A common floor η implies $\sum U_0 \ge 3\eta$ and $2^L \sum U_0 \le 3(1 - \eta)$, so $\eta \le 1/(2^L + 1)$. A ceiling M implies $\sum U_0 \ge 3(1 - M)$ and $2^L \sum U_0 \le 3M$, so $M \ge 2^L/(2^L + 1)$. Both optima are unique at $U_{\ell i} = 2^\ell/(2^L + 1)$. Thus $x = z$ although a top V has $x_i/H_i = 1/(2^L + 1)$.

8 Separation from previously studied subclasses

The recurrence array has genuine three-way interaction. Take sector coordinates $A = (G_0, U_{00})$, $B = (G_1, V_{00})$, $C = (G_2, D_{00})$ and index the eight corners by zeros and ones. Only $w_{111} = 2$, $w_{110} = 2$ and $w_{100} = 1$ are nonzero. Its third mixed difference is

$$w_{111} - w_{110} - w_{101} - w_{011} + w_{100} + w_{010} + w_{001} - w_{000} = 1.$$

Every decomposition $w_{abc} = f_{ab} + g_{ac} + h_{bc}$ gives zero by cancellation. Thus the family is outside even the pair-additive class, irrespective of relabeling; it is not a pair market represented using decorative dummy agents.

A recent result for supermodular multisided arrays satisfying local monotonicity gives simultaneous attainment of all coordinate maxima within any one sector [11, Theorem 3.2]. Our core violates this property for $L \geq 2$. Since $(J - I)^2 = J + I$, every entry of $(J - I)^L$ is strictly positive. A top V attains its maximum one only at $q_0 = 0$, while a root U in the same sector has maximum greater than one, as witnessed by a small positive feasible q_0. Every sector contains both types. Hence no ordering of this array satisfies that theorem's hypotheses. These are explicit exclusions of inspected subclasses, not a proof of priority against all possible representations.

9 Priority and interpretation

What is already established. Core/optimal-dual descriptions, matched-team complementarity, and the multisided model are established background [8, 9]. Sorted egalitarian selection is not new [5, 4]. Feasible pairwise equal splits are Lorenz-best, and exact assignment games have coincident selections [6, 7]. These facts are not candidate contributions here. Both extremal families are nonexact TU games: a distinguished player has core minimum one despite singleton worth zero.

A direct dependence discovered in this audit. For the bipartite half-guarantee, start Vazirani's Algorithm 1 from an arbitrary core point y in its rational-weight domain. Its clock starts at the minimum initial essential payoff $\Omega_0 \geq 0$; nonfixed payoffs move at speed at most one, and upon permanent freezing at clock value t_i the final payoff obeys $x_i \geq t_i$. The update, freezing and floor invariants imply

$$|x_i - y_i| \leq t_i - \Omega_0 \leq x_i - \Omega_0, \qquad 2x_i \geq y_i + \Omega_0 \geq y_i.$$

Initially fixed coordinates are unchanged. Relevant anchors in [4] are Definition 19, Algorithm 1, Remark 1, Lemmas 5 and 9, and Theorem 8 (printed pp. 17–18, 20, 22, 25, 27, 29–30). The inequality is our deduction from these statements, not a located explicit theorem in that source. Accordingly the bipartite half bound is best positioned as a short corollary with an elementary proof, not as a wholly independent fairness discovery.

The general-graph extension also follows by the classical double-cover reduction [1]. Duplicate every vertex and replace each edge ij by the two crossed edges $i'j''$ and $j'i''$ of the same weight. Duplicating an optimal matching gives worth at least $2T$; duplicating a core vector gives a feasible cover of total $2T$, proving equality. Uniqueness makes the bipartite leximin vector invariant under the global copy swap, so it is (x, x). Symmetric core vectors are exactly duplicated original core vectors, and duplication preserves sorted comparisons. Applying the bipartite half guarantee to every duplicated original core point gives the graph half guarantee. This is a short inference from the known reduction, not a separately substantial new extension. It does not give the sharp finite bound: the matching parameter doubles from k to $2k$.

The additive sector-shift core is also established geometry: complete nonnegative additive arrays are a special case of the Böhm–Bawerk core representation [2]. Its 1/3 and 4/3 constants here are elementary quantitative benchmarks.

Candidate contribution and limits. The sharp finite distance bound, its attainment, the sparse bounded-integer three-sided recurrence, and the combined comparison are the main candidates. No equivalent quantitative theorem was located in the dated source audit. That statement does not certify priority: original Quint and Lucas full texts remain unavailable, as does the 2023 preprint on extreme multisided core points [3]. The preserved audit records exact access and citation-chain coverage. Existing assortative and local-monotonicity results impose extra hypotheses; a three-way transportation-polytope representation also would not automatically preserve player coordinates and full selectors.

The economic content is a worst-case dependence on nonadditive three-way cooperation. The entire core has only three degrees of freedom. The construction uses forced-zero dummies, exact constraints and replicated top pairs. It does not show that these extremes are common in observed markets or robust to arbitrary perturbations. The additive result isolates a simple protected subclass; it makes no assertion about every assortative market. The intended contribution is this sharp worst-case comparison, rather than an empirical claim about matching markets. Independent specialist review of correctness and priority is pending.

Verification and provenance. The accompanying portable package regenerates exhaustive bounded 2×2 and $2 \times 2 \times 2$ calibrations, sampled $3 \times 3 \times 3$ controls, and general-graph and additive checks. It also includes independent sorted-rank improvement queries and a separate raw-triple construction through 192 players. Run `python verification/run_checks.py`; the README records dependencies, coverage and limitations. All-parameter conclusions rest on the proofs above, not on finite enumeration. The claim/source ledger distinguishes established results, our deductions, and unresolved priority questions. Codex assisted with research development, drafting and computational verification; the provenance file records source versions and package changes. External specialist correctness and priority review is pending.

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